

Lecture 3: Physical Layer

Guided Media and Digital Signaling

EC 441 - Introduction to Computer Networking
Boston University

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Learning Objectives

By the end of this lecture, you should be able to:

- Describe characteristics of guided transmission media
- Calculate signal attenuation using dB/km models
- Understand relationship between bandwidth and data rate
- Model communication channels using AWGN
- Analyze digital signaling schemes
- Design and analyze matched filter receivers
- Calculate probability of error using Q-function
- Understand tradeoffs: power, bandwidth, error rate, data rate

Why Study Physical Media?

Networks transmit **electromagnetic energy**:

- Electrical signals on copper wires
- Light pulses in optical fibers
- Radio waves through air

Understanding constraints is critical for:

- Link budget analysis (can signal reach destination?)
- Data rate limits (how fast can we transmit?)
- Network design (which medium for which application?)

Guided Media Comparison

Medium	Bandwidth	Distance	Cost	Use Case
Twisted Pair	1-1000 MHz	100 m	Low	Ethernet, phone
Coaxial	1-1000 MHz	500 m	Medium	Cable TV
Optical Fiber	>10 THz	100 km+	High	Backbone

Key insight: Each medium optimized for different applications

Twisted Pair Cable

Structure:

- Two insulated copper wires twisted together
- Twisting reduces electromagnetic interference (EMI)
- Multiple pairs bundled in cable

Categories (Ethernet):

- Cat 5e: 100 MHz, up to 1 Gbps
- Cat 6: 250 MHz, up to 10 Gbps (short distance)
- Cat 6a: 500 MHz, 10 Gbps up to 100 m
- Cat 8: 2000 MHz, 25-40 Gbps (data centers)

Twisted Pair: Key Parameters

Characteristic Impedance: $Z_0 = 100 \, \Omega$ (for Ethernet)

- Must match transmitter/receiver to avoid reflections

Attenuation: $\sim 2\text{-}3 \text{ dB per } 100\text{m at } 100 \text{ MHz}$

- Increases with frequency: $\alpha \propto \sqrt{f}$

Propagation speed: $v \approx 0.64c$ (where c = speed of light)

- Dielectric slows signal compared to free space

Twisted Pair: Example

Cat 6 cable at 100 MHz over 100m:

- Attenuation: ~ 2.5 dB/100m
- Transmit power: $P_{\text{tx}} = 1 \text{ W} = 30 \text{ dBm}$
- Received power: $P_{\text{rx}} = 30 - 2.5 = 27.5 \text{ dBm} \approx 0.56 \text{ W}$
- Power loss: 44%

Note: Moderate loss, suitable for short distances

Coaxial Cable

Structure:

- Central conductor (copper wire)
- Dielectric insulator
- Outer conductor (shield/braid)
- Protective jacket

Types:

- RG-6 (75 Ω): Cable TV, cable modem
- RG-58 (50 Ω): Legacy Ethernet

Key Parameters:

- Attenuation: 0.5-2 dB per 100m (lower than twisted pair)
- Bandwidth: DC to 1+ GHz
- Modern use: Cable Internet (DOCSIS 3.1, up to 10 Gbps)

Basic Principle:

- Light propagates via **total internal reflection**
- Core has higher refractive index than cladding
- Critical angle: $\theta_c = \arcsin(n_2/n_1)$

Types:

Single-Mode (SMF):

- Core: $\sim 9 \mu\text{m}$
- One light path
- Distance: 100+ km
- Use: Long-haul

Multi-Mode (MMF):

- Core: 50-62.5 μm
- Multiple paths
- Distance: 2 km
- Use: Data centers

Optical Fiber: Key Parameters

Attenuation (ultra-low!):

- 1550 nm: ~ 0.2 dB/km
- 1310 nm: ~ 0.4 dB/km
- 850 nm (MMF): ~ 2.5 dB/km
- Compare to copper: ~ 25 dB/km at 100 MHz
- **Fiber has $100\times$ less attenuation!**

Bandwidth: Enormous!

- Single fiber: >10 THz spectrum
- WDM: $100+$ wavelengths $\times$ 100 Gbps = $10+$ Tbps/fiber

Propagation speed: $v \approx 0.67c$ (in glass, $n \approx 1.5$)

Fiber Example: Long-Haul Link

100 km fiber link at 1550 nm:

$$\text{Attenuation} = 0.2 \text{ dB/km} \times 100 \text{ km} = 20 \text{ dB}$$

$$P_{\text{tx}} = 1 \text{ mW} = 0 \text{ dBm}$$

$$P_{\text{rx}} = 0 - 20 = -20 \text{ dBm} = 10 \mu\text{W}$$

Power ratio = 100 : 1 loss (still plenty of signal!)

This is why fiber dominates long-distance communications

Decibels (dB)

Why use dB? Multiplication becomes addition!

Power ratio in dB:

$$\text{dB} = 10 \log_{10} \left(\frac{P_{\text{out}}}{P_{\text{in}}} \right)$$

Absolute power levels:

- **dBm** (relative to 1 mW): $\text{dBm} = 10 \log_{10} \left(\frac{P}{1 \text{ mW}} \right)$
- **dBW** (relative to 1 W): $\text{dBW} = 10 \log_{10} \left(\frac{P}{1 \text{ W}} \right)$

Common dB Values to Memorize

Power Ratio	dB	Meaning
2	3 dB	Double
10	10 dB	10×
100	20 dB	100×
1/2	-3 dB	Half
1/10	-10 dB	0.1×

Link Budget:

$$P_{\text{rx}}(\text{dBm}) = P_{\text{tx}}(\text{dBm}) - L_{\text{cable}}(\text{dB}) + G_{\text{tx}} + G_{\text{rx}}$$

Shannon-Hartley Theorem

$$C = B \log_2 \left(1 + \frac{S}{N} \right) \text{ bits/s}$$

where:

- C = channel capacity (max achievable data rate)
- B = bandwidth (Hz)
- S/N = signal-to-noise ratio (linear, not dB)

Key Insights:

- 1 Capacity $\propto$ bandwidth (linear)
- 2 Capacity $\propto \log(\text{SNR})$ (diminishing returns)
- 3 **Bandwidth is precious!**
- 4 Can trade SNR for bandwidth

Shannon Capacity Examples

Telephone channel: $B = 3$ kHz, $\text{SNR} = 30$ dB (1000:1)

$$C = 3000 \times \log_2(1001) \approx 30 \text{ kbps}$$

Ethernet link: $B = 100$ MHz, $\text{SNR} = 20$ dB (100:1)

$$C = 10^8 \times \log_2(101) \approx 666 \text{ Mbps}$$

Note: Practical systems achieve 50-90% of Shannon capacity

Propagation Speed

Speed of EM waves in medium:

$$v = \frac{c}{\sqrt{\epsilon_r}}$$

Typical Values:

- Vacuum: $v = c = 3 \times 10^8$ m/s
- Air: $v \approx 0.997c$
- Twisted pair: $v \approx 0.64c$ ($\epsilon_r \approx 2.3$)
- Coax: $v \approx 0.66c$
- Optical fiber: $v \approx 0.67c$ ($n \approx 1.5$)

Example: NY to LA (~ 4000 km)

- Fiber propagation: 20 ms one-way
- Round-trip: ~ 40 ms (propagation only)
- Measured ping: 60-80 ms (includes processing)

Communication System Model

[Encoder] → [Modulator] → [Channel] → [Demodulator] → [Decoder]

Simplified physical layer model:

- Transmitter generates signal $x(t)$
- Channel modifies signal (attenuation, delay)
- Noise adds random disturbance $n(t)$
- Receiver observes $y(t) = h(t) * x(t) + n(t)$

Mathematical Model:

$$y(t) = h(t) * x(t) + n(t)$$

Noise Characteristics:

- ① **Additive:** Noise adds to signal
- ② **White:** Equal power at all frequencies
 - Power spectral density: $N_0/2$ W/Hz
- ③ **Gaussian:** Amplitude follows Gaussian distribution
 - $n(t) \sim \mathcal{N}(0, \sigma^2)$

Why AWGN?

- Mathematically tractable
- Good model for thermal noise
- Fundamental performance limit

Thermal Noise

Johnson-Nyquist noise: Random electron motion in conductors

Available noise power:

$$N = k_B T B$$

where:

- $k_B = 1.38 \times 10^{-23}$ J/K (Boltzmann constant)
- T = temperature (Kelvin)
- B = bandwidth (Hz)

Example: Room temp (300 K), 1 MHz bandwidth

$$N = 1.38 \times 10^{-23} \times 300 \times 10^6 = 4.14 \times 10^{-15} \text{ W} = -144 \text{ dBm}$$

Key insight: Noise power $\propto$ bandwidth!

Noise Figure

Noise Figure (NF): Measures SNR degradation by component

$$\text{NF (dB)} = 10 \log_{10} \left(\frac{\text{SNR}_{\text{in}}}{\text{SNR}_{\text{out}}} \right)$$

Friis Formula (cascaded components):

$$F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots$$

Key insight: First stage (LNA) dominates!

- Keep first-stage NF as low as possible
- High first-stage gain reduces impact of later stages

Signal-to-Noise Ratio (SNR)

Definition:

$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \frac{S}{N}$$

In dB:

$$\text{SNR (dB)} = 10 \log_{10} \left(\frac{S}{N} \right) = P_s(\text{dBm}) - P_n(\text{dBm})$$

Typical Values:

- Voice (acceptable): 10-15 dB
- Music (good quality): 40-60 dB
- Digital communication: 10-30 dB (depends on modulation)

Representing Bits as Signals

Goal: Transmit digital data (bits) over physical channel

General transmitted signal:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT)$$

where:

- a_k = symbol values (from data bits)
- $g(t)$ = pulse shape (determines bandwidth)
- T = symbol period (symbol rate = $1/T$)

Key Design Choices:

- 1 Pulse shape $g(t)$
- 2 Symbol mapping a_k
- 3 Symbol rate $1/T$

Unipolar (On-Off Keying):

- Bit 0: $a_k = 0$
- Bit 1: $a_k = A$
- Simple, but wastes energy

Polar (Bipolar) NRZ:

- Bit 0: $a_k = -A$
- Bit 1: $a_k = +A$
- Balanced (no DC component)
- Better noise immunity
- **Most common for baseband**

Pulse Shapes

1. Rectangular Pulse:

$$g(t) = \begin{cases} 1 & 0 \leq t < T \\ 0 & \text{otherwise} \end{cases}$$

- Bandwidth: $B \approx 1/T$
- Simple to generate
- Causes intersymbol interference

2. Raised Cosine Pulse:

- Bandwidth: $B = \frac{1+\alpha}{2T}$ (α = roll-off factor)
- Zero ISI at sampling points
- Controllable bandwidth
- Used in WiFi, cellular, satellite

Energy per Bit and E_b/N_0

Energy per bit:

$$E_b = \int_0^T x^2(t) dt$$

For bipolar NRZ: $E_b = A^2 T$

Key performance metric: E_b/N_0

$$\frac{E_b}{N_0} = \frac{\text{Energy per bit}}{\text{Noise spectral density}}$$

Why use E_b/N_0 instead of SNR?

- Normalizes for data rate
- Independent of bandwidth
- Universal metric across modulation schemes

Optimal Receiver Design

Goal: Minimize bit error probability in AWGN

Received signal:

$$y(t) = a_k g(t - kT) + n(t)$$

Matched Filter Theorem:

To maximize SNR at sampling instant, use filter matched to pulse:

$$h_{\text{MF}}(t) = g(T - t)$$

Receiver Structure:

$$y(t) \rightarrow [\text{Matched Filter}] \rightarrow [\text{Sample at } t=T] \rightarrow [\text{Decide}] \rightarrow \hat{a}_k$$

For binary signaling ($a_k \in \{-A, +A\}$):

Decision rule:

- If $z > 0$: decide $\hat{a}_k = +A$ (bit 1)
- If $z < 0$: decide $\hat{a}_k = -A$ (bit 0)

This is the **maximum likelihood (ML) decision** for equally likely bits.

Threshold:

- Symmetric signaling (bipolar): threshold = 0
- Asymmetric (OOK): threshold = $(A_0 + A_1)/2$

Probability of Error

Received sample after matched filter:

$$z = a_k + n_s$$

where $n_s \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = N_0/2$

For bipolar NRZ:

Error if noise pushes sample across threshold:

$$P(\text{error}) = Q\left(\frac{A}{\sigma}\right)$$

Bit Error Rate (BER):

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

The Q-Function

Definition:

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-u^2/2} du$$

Key property: Decreases **very rapidly** with x

- $Q(3) \approx 1.3 \times 10^{-3}$ (1 error per 1000 bits)
- $Q(4) \approx 3 \times 10^{-5}$ (1 per 30,000)
- $Q(5) \approx 3 \times 10^{-7}$ (1 per 3 million)
- $Q(6) \approx 10^{-9}$ (1 per billion)

Waterfall effect: Small SNR increase $\Rightarrow$ huge BER decrease!

BER vs. E_b/N_0

E_b/N_0 (dB)	Argument	BER
6 dB	$Q(2)$	2.3×10^{-2}
8 dB	$Q(2.52)$	6×10^{-3}
10 dB	$Q(3.16)$	8×10^{-4}
12 dB	$Q(4)$	3×10^{-5}
15 dB	$Q(5.6)$	10^{-8}

Typical targets: $\text{BER} < 10^{-6}$ to 10^{-9}

M-ary PAM

Motivation: Send more bits per symbol $\Rightarrow$ higher data rate

M-ary Pulse Amplitude Modulation:

- $M = 2$: Binary (1 bit/symbol)
- $M = 4$: 4-PAM (2 bits/symbol)
- $M = 8$: 8-PAM (3 bits/symbol)
- $M = 16$: 16-PAM (4 bits/symbol)

Example: 4-PAM

- Bits 00: $a_k = -3A$
- Bits 01: $a_k = -A$
- Bits 10: $a_k = +A$
- Bits 11: $a_k = +3A$

Multi-Level PAM: Performance

Minimum distance: $d_{\min} = 2A$

Probability of symbol error:

$$P_s \approx 2 \left(1 - \frac{1}{M}\right) Q\left(\frac{d_{\min}}{2\sigma}\right)$$

Key insight: Doubling M requires ~ 6 dB more E_b/N_0 for same BER!

Tradeoff:

- Higher $M \Rightarrow$ better bandwidth efficiency
- But requires more power (higher SNR)

Bandwidth Efficiency

Symbol rate: $R_s = 1/T$ symbols/sec

Bit rate: $R_b = R_s \log_2(M)$ bits/sec

Spectral efficiency: $\eta = R_b/B$ (bits/s/Hz)

M	bits/symbol	η (bits/s/Hz)
2	1	1.33
4	2	2.67
8	3	4.0
16	4	5.33

(Assumes raised cosine with $\alpha = 0.5$)

Fundamental Tradeoffs

1. Power vs. Data Rate (fixed bandwidth):

- Higher data rate requires more power
- Shannon: $C = B \log_2(1 + \text{SNR})$

2. Bandwidth vs. Data Rate (fixed power):

- Higher data rate requires more bandwidth

3. Error Rate vs. SNR:

- Lower BER requires higher E_b/N_0
- Exponential improvement (Q-function)

4. Multi-level signaling:

- More bits/symbol $\Rightarrow$ better bandwidth efficiency
- But requires higher SNR for same BER

Key Takeaways

- ① **Physical media:** Fiber dominates long-distance (low loss, huge bandwidth)
- ② **Link budgets:** Use dB for calculations, account for margins
- ③ **Shannon capacity:** $C = B \log_2(1 + \text{SNR})$
 - Linear in bandwidth, logarithmic in power
- ④ **AWGN model:** Thermal noise sets fundamental limit
- ⑤ **Digital signaling:** Bipolar NRZ most common
- ⑥ **Matched filter:** Maximizes SNR at sampling instant
- ⑦ **BER:** Waterfall effect - small SNR gain $\Rightarrow$ huge BER drop
- ⑧ **Multi-level PAM:** Bandwidth vs. power tradeoff

Physical limits constrain network design:

- Ethernet limited to 100m (timing, not attenuation)
- Fiber enables long-haul (1000s km)
- Wireless has severe power constraints

Data rates tied to SNR and bandwidth:

- WiFi adapts modulation based on SNR
- Higher rates need better signal quality

Error rates affect protocols:

- Wired: BER $\sim 10^{-9}$, minimal retransmission
- Wireless: BER $\sim 10^{-5}$ to 10^{-3} , strong FEC needed

Lecture 4: Physical Layer - Wireless Communications

- Transmitter and receiver architectures
- Propagation models (path loss, fading)
- Modulation as channel selection (QAM, OFDM)
- Spectrum allocation
- Cellular reuse patterns